



Ingredient Recommendation System

food.com - Consulting Project

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Agenda

01



Project brief

What food.com asked us to build

02

Data & EDA

Ingredient space, ratings, calories

03



System design

Three capabilities, one co-occurrence model

04



Results & evaluation

Precision@K, Recall@K, coverage, diversity

05



Limitations & next steps

What we'd improve with more time

What food.com Asked Us to Build

Three required deliverables

- 1 Recommend the Top-N ingredients a user is most likely to want, based on past recipe reviews
- 2 Given a list of ingredients, return the best-matching recipes from the catalogue
- 3 Allow users to replace a specific ingredient with a relevant culinary alternative

Optional objectives

Coverage & diversity

Measure how broadly the system explores the 14,942 unique ingredients

Calorie-aware ranking

Incorporate nutrition data to promote healthier recommendations

Data at a Glance

231K

Recipes
in metadata.csv

55K

Unique users
in train.csv

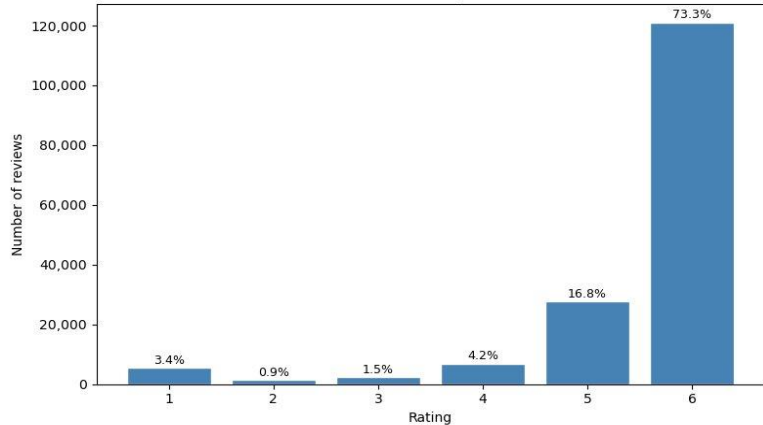
197K

User-recipe
interactions

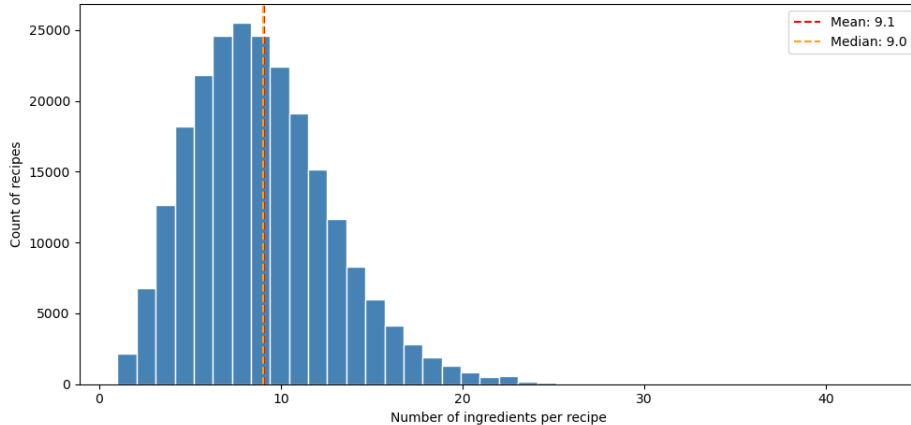
14,942

Unique
ingredients

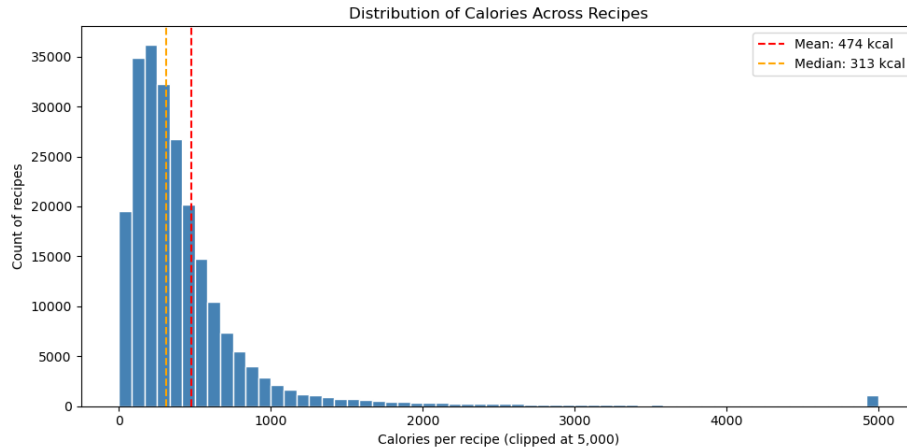
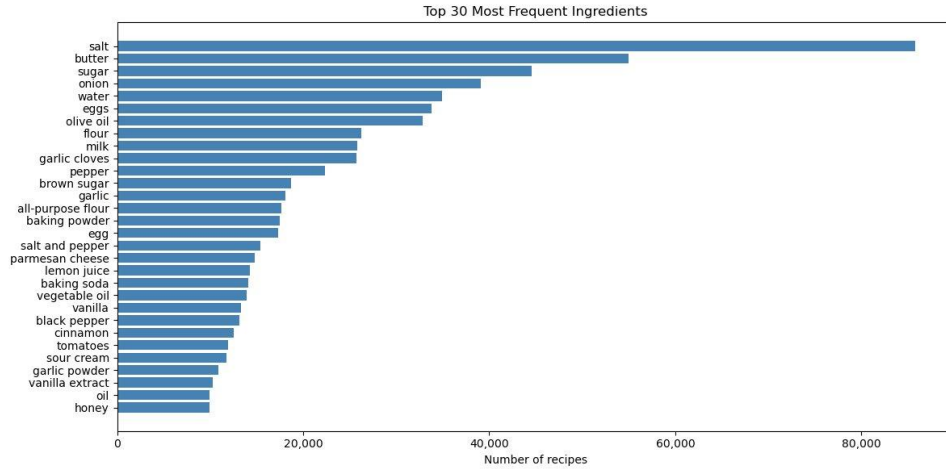
Distribution of Ratings in train.csv



Distribution of Ingredients per Recipe



EDA - Ingredient Frequency & Calorie Profile



Key insights

Salt alone appears in 85K+ recipes - ubiquitous staples dominate and will bias a naive model.

Median calorie: 313 kcal - used as the healthiness threshold for calorie-aware re-ranking.

Data Preparation: From Ratings to Ingredient Signals



User-level split: all validation and test users are cold-start - the realistic deployment scenario

Split	Users	Proportion
Train	~38,500	70%
Validation	~8,250	15%
Test	~8,250	15%

Cold-start check

Ingredient cold-start rate $< 1\%$

Most common ingredients span all users. A global popularity fallback handles the rare unseen cases.

System Design

One co-occurrence matrix - three capabilities

Capability 1 - Top-N Ingredient Recommendation

How it works

1. Build co-occurrence matrix

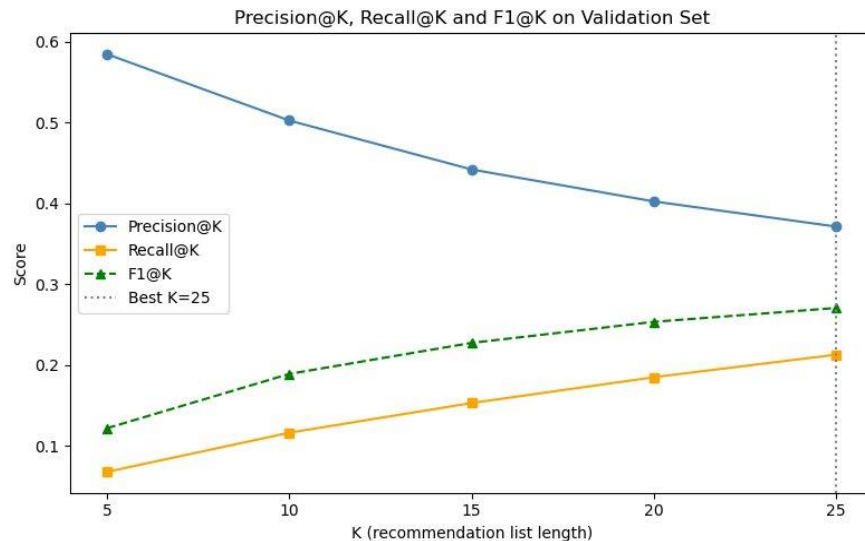
Count ingredient pairs across all 231K recipes

2. Score unseen ingredients

Sum co-occurrence counts with the user's seed ingredients

3. Rank and return Top-K

Popularity fallback for cold-start cases



K=10 chosen as operating value - F1 peaks at K=25 but a 25-item list is impractical for a food app. K=10 delivers Precision@10 = 0.503: one in two recommendations is relevant, even for cold-start users.

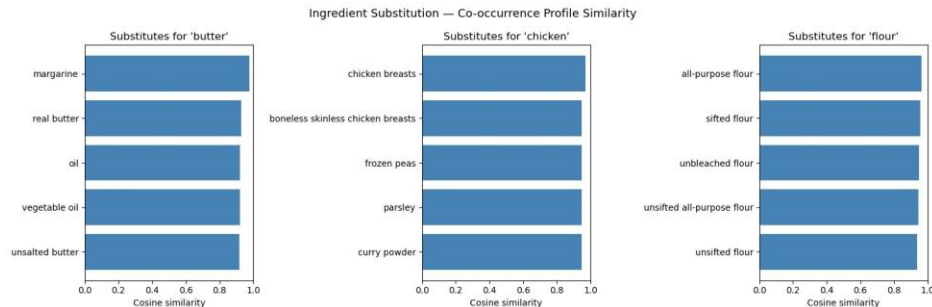
Capability 2 - Recipe Matching

Recipe Matching

Given the user's Top-10 ingredient list, retrieve recipes scored by:

- Overlap count - matched ingredients
 - Jaccard similarity - normalises for recipe length
- Calories displayed to support health objective.

Capability 3 - Ingredient Substitution



Method: cosine similarity between co-occurrence profile vectors. Ingredients with similar profiles appear in the same recipe types - they play the same culinary role.

butter → *margarine* (0.977), *oil* (0.920) • *flour* → *all-purpose flour* (0.964)

All three capabilities are built on a single co-occurrence matrix - no additional training data or models required.

Evaluation - Why Precision@K and Recall@K, Not RMSE

This is a Top-N ranking problem, not a rating prediction problem



Precision@K

Fraction of the K recommendations that are genuinely relevant.

Measures list quality as the user experiences it.



Recall@K

Fraction of all relevant items that appear in the top K.

Measures coverage of the user's true preferences.



F1@K

Harmonic mean of Precision and Recall.

Used for hyperparameter selection (K tuning on the validation set).

RMSE / MAE are not appropriate here - they require scalar rating predictions vs ground-truth ratings. Our system outputs ranked lists of unrated candidates.

Test Set Results

0.585

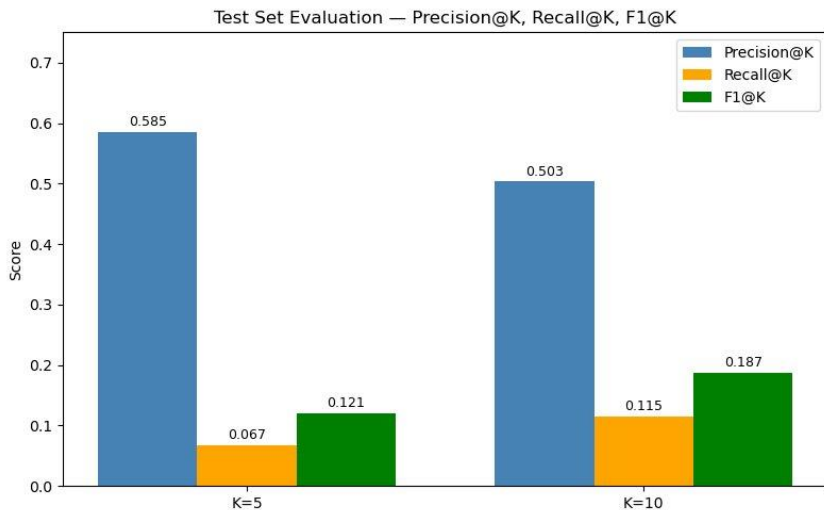
Precision@5
(test set)

0.503

Precision@10
(test set)

0.001

Val → Test gap in P@10
(consistent generalisation)



What these numbers mean

1 in 2 recommended ingredients is genuinely relevant - strong for cold-start users.

Low recall is expected: users have ~90 ingredient preferences. A list of 10 covers only ~12% of that space.

Val → Test gap of 0.001 confirms the model generalises consistently.

Optional Objective 1 - Coverage & Diversity



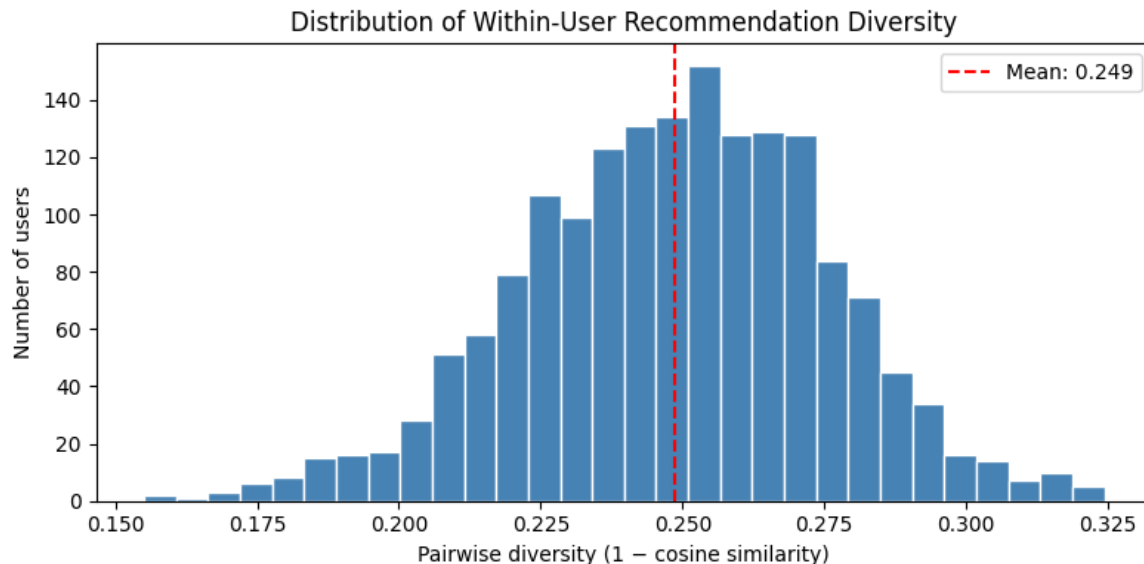
Coverage: 0.27% (41 of 14,942 ingredients ever recommended)

Co-occurrence scoring concentrates on popular pantry staples. Long-tail ingredients - niche, regional - are never surfaced. This is the model's most important weakness.



Within-user diversity: 0.249 / 1.0

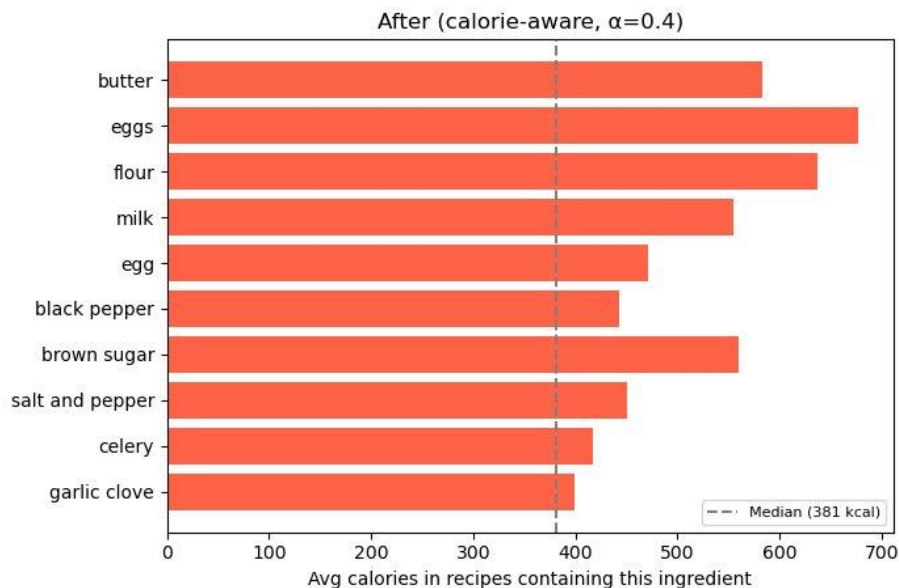
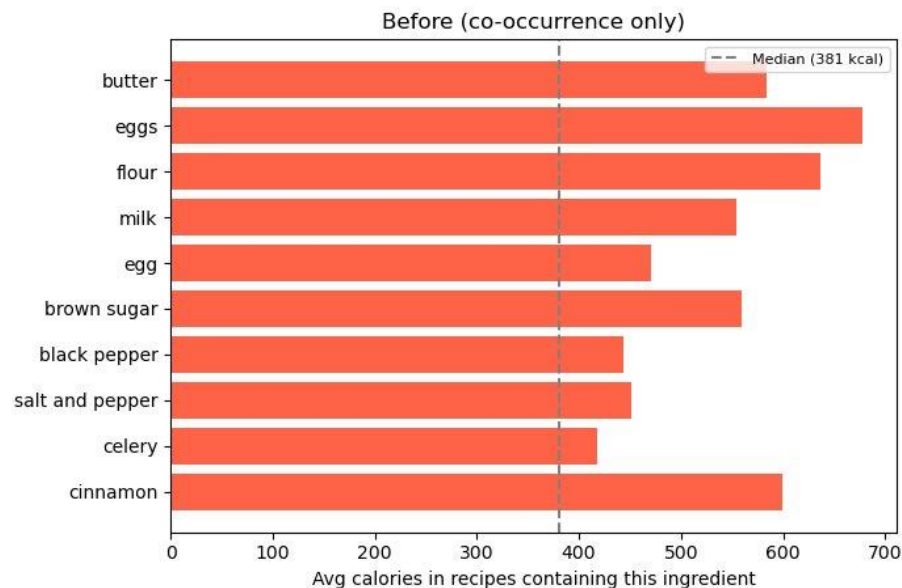
Average pairwise dissimilarity within a Top-10 list. Moderate variety - not all identical - but lists trend toward similar culinary roles.



Optional Objective 2 - Calorie-Aware Re-ranking

Score formula: $normalised_cooccurrence - 0.4 \times normalised_calorie$ ($\alpha = 0.4$ controls calorie penalty strength)

Calorie-aware Re-ranking — User U7411927



Result: avg calorie association drops ~47 kcal per recommendation. Cinnamon (above-median) replaced by garlic clove (below-median). **Limitation:** calorie values reflect the surrounding dish, not the ingredient itself. All 7 nutrition dimensions (protein, fat, sugar, sodium...) should be weighted in production.

Limitations & Future Work

Coverage: 0.27% - the biggest weakness



Co-occurrence scoring is proportional to ingredient frequency. Fix: Maximum Marginal Relevance (MMR), category constraints, or SVD on the user-ingredient matrix to learn personalised embeddings for all 14,942 ingredients.

Evaluation split is effectively random



Ingredients have no timestamps at the individual level, so the 50/50 seed split is unordered. A time-ordered split using recipe review dates would better simulate deployment and give more realistic performance estimates.

Substitution cannot separate substitutes from complements



Butter and flour have similar co-occurrence profiles (both appear in baking) but they complement each other, not substitute. Ingredient category metadata (USDA food groups) would constrain candidates to the same food category.

Conclusion



0.503

Precision@10 on cold-start test users

~1 in 2 recommendations relevant



-47 kcal

Avg calorie reduction per recommendation

Calorie-aware re-ranking ($\alpha=0.4$)



0.27%

Catalogue coverage - biggest weakness

Only 41 of 14,942 ingredients ever recommended



SVD

Recommended next step

Matrix factorisation on user-ingredient matrix



Thank You
For Listening!

